Equazioni di Maxwell

2/2/21 5:38 PM

VUOTO

I: $\nabla \cdot \bar{E} = \frac{\rho}{\varepsilon_0}$ IV $\nabla \cdot B = 0$

II: $\bar{\nabla} \times \bar{E} = -\frac{\partial \bar{B}}{\partial t}$ IV $\bar{\nabla} \times \bar{B} = \mu_0 J + \mu_0 \varepsilon_0 \frac{\partial \bar{E}}{\partial t} = \mu_0 (J + J_s)$

IN UN MEZZO

$\bar{D} = \varepsilon_0 \bar{E} + P$ $\bar{H} = \frac{\bar{B}}{\mu_0} - \bar{M}$

I: $\nabla \cdot D = \rho$ II: $\nabla \cdot B = 0$

III: $\bar{\nabla} \times \bar{E} = -\frac{\partial B}{\partial t}$ IV: $\bar{\nabla} \times \bar{H} = \bar{J} + \frac{\partial D}{\partial t} = \bar{J} + \bar{J}_c$

Se lineare

$\bar{D} = \underbrace{\varepsilon_0 \varepsilon_r}_{\varepsilon} \bar{E}$ $\bar{B} = \underbrace{\mu_0 \mu_r}_{\mu} \bar{H}$